Assessing HAB Risk using Scripps Institute Data

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Emails

UC San Diego
HALICIOĞLU DATA SCIENCE INSTITUTE

Abstract

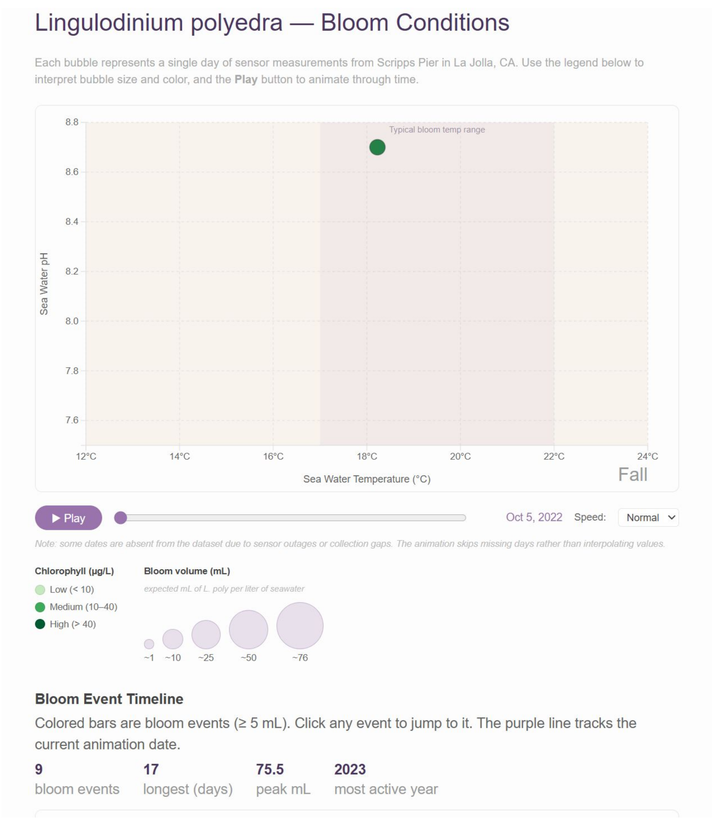
We studied harmful algal blooms caused by *Lingulodinium polyedra* using Imaging FlowCytobot (IFCB) and environmental data from Scripps Pier. Machine learning models were used to predict bloom abundance and analyze seasonal patterns. Our work supports improved monitoring and early warning of HAB events.

Background

The *Lingulodinium polyedra* has shown to cause red tides and produces harmful toxins that directly affect marine life and create adverse effects on human health. Previous studies show that *L. polyedra* blooms are influenced by temperature, salinity, nutrient availability, water flow, and seasonal variability. Bloom events are most common during warmer months and under favorable oceanographic conditions. With the data from Scripps Institute using the IFCB we conducted our own study

Objectives We Answered

- We used ifcb data to predict the abundance of L .poleydra
- Identified environmental factors associated with bloom patterns and bloom
- Build and deploy an interactive visualization platform to depict relationships with environmental factors and L. polyedra.
- Construct and provide analysis on bloom formation for both image and environmental features.



Results and Conclusion

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